

**An Internship Report**  
Of  
**Data Science Master Virtual Internship**

Submitted  
To  
**School of Computer Science and Engineering**  
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
**B.Tech (Computer Science and Engineering )**



By  
Roll Number of Student: 2410030458  
Name of Student: Adyasha Das  
Batch: 2024-2028

## **CANDIDATE'S DECLARATION**

I, Adyasha Das, do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: Adyasha Das

Roll No.: 2410030458

## **ACKNOWLEDGEMENT**

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, Altair RapidMiner Academy, and Siemens for providing me the opportunity and the structured platform to learn and gain practical exposure in the domain of Data Science and Machine Learning. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: Adyasha Das

Roll No.: 2410030458

| <b>TABLE OF CONTENTS</b>                                      | <b>Pg.No</b> |
|---------------------------------------------------------------|--------------|
| 1. Candidate's Declaration .....                              | 2            |
| 2. Acknowledgement .....                                      | 3            |
| 3. Internship Completion Certificate .....                    | 5            |
| 4. Internship Offer Letter.....                               | 6            |
| 5. Project Description .....                                  | 8            |
| 5.1 Introduction .....                                        | 8            |
| 5.2 Organization Profile .....                                | 8            |
| 5.3 Problem Statement .....                                   | 8            |
| 5.4 Project Objectives .....                                  | 8            |
| 5.5 Scope of the Project .....                                | 9            |
| 5.6 Technologies and Tools Used .....                         | 9            |
| 5.7 System Architecture .....                                 | 9            |
| 5.8 Methodology .....                                         | 10           |
| 5.9 Expected Outcomes .....                                   | 11           |
| 5.10 Certificates of Completion and Communication Proof ..... | 11           |
| 6. Bibliography/References .....                              | 13           |

### 3. INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय  
MINISTRY OF  
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्  
All India Council for Technical Education



NATIONAL  
INTERNSHIP  
PORTAL



EduSkills®  
Nation Building Through Skills



## Certificate of Virtual Internship

This is to certify that

**Adyasha Das**

**IILM University, Greater Noida**

has successfully completed the 8-weeks

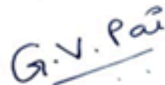
**Data Science Master Virtual Internship**

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

**SIEMENS**



Gaurav Pai  
Head - Strategy and Operations  
Siemens Industry Software (India) Pvt. Ltd



Dr. Buddha Chandrasekhar  
Chief Coordinating Officer (CCO)  
AICTE, Ministry of Education



Shubhajit Jagadev  
Chief Executive Officer (CEO)  
EduSkills



Certificate ID : 4416935e55146b2e7ece  
Student ID: STU6986f3c62cb241770451910



GRADE- O (Outstanding)-90-100 | E (Excellent)-80-89 | A (Very Good)-70-79 | B (Good)-60-69 | C (Fair)-50-59 | D (Average)-40-49 | P (Pass)-30-39 | F (Fail)-Below 30

## 4. Internship Offer Letter



Ref No: C17Y2026M081497680

### Internship Offer Letter

**Student Name** : Adyasha Das  
**Stream** : B.Tech  
**Branch** : B.Tech(Computer Science and Engineering)  
**Institute Name** : IILM University, Greater Noida  
**Internship Domain** : Siemens Data Science Master  
**Internship Duration** : June 2026 - August 2026  
**AICTE Student ID** : STU6986f3c62cb241770451910

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

#### **Internship Structure & Evaluation:**

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

## Internship Offer Letter — Continued



**The week wise Structure of the Internship is as Follows:**

| Week                                                            | Module                                | Description                                                               |
|-----------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------|
| Week 1                                                          | Applications & Use Cases Professional | Learn to identify business problems suitable for AI and machine learning. |
| Week 2                                                          | Data Engineering Professional         | Learn to access, load, transform, and calculate data.                     |
| Week 3                                                          | Machine Learning Professional         | Build foundational machine learning skills.                               |
| Week 4                                                          | Data Preparation Master               | Master advanced data handling and automation.                             |
| Week 5                                                          | Machine Learning Master               | Learn advanced machine learning techniques.                               |
| Week 6                                                          | Applications & Use Cases Master       | Deploy and monitor ML models.                                             |
| Week 7                                                          | Platform Administration Master        | Configure and manage RapidMiner AI Hub.                                   |
| Week 8                                                          | Platform Administration Master        | Enterprise-level platform operations.                                     |
| Final Credential Validation / Internship Grade Point Assessment |                                       |                                                                           |

**Note: There is no stipend associated with this internship.**

We congratulate you on your selection and wish you a successful learning journey.



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## **5. PROJECT DESCRIPTION**

### **5.1 Introduction**

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “Siemens Data Science Master”. The internship was carried out from June 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to build a strong practical foundation in Data Science, Machine Learning and Artificial Intelligence using RapidMiner, and to develop an end-to-end understanding of data preparation, machine-learning workflows, model validation, deployment and platform administration.

The internship followed a structured, week-wise curriculum combining conceptual learning with practical learning-path activities, quizzes, assessments and certification. On successful completion of the required activities and Final Credential Validation / Internship Grade Point Assessment, a Virtual Internship Certificate with an “Outstanding” grade was awarded.

### **5.2 Organization Profile**

EduSkills Academy is an organization working under the theme “Nation Building Through Skills” and partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal.

For the Siemens Data Science Master track, EduSkills provides a structured learning pathway covering Data Science, Data Engineering, Machine Learning, Applications & Use Cases, Data Preparation and Platform Administration. The technical learning is supported through RapidMiner learning paths and assessments designed to develop practical and industry-relevant Data Science capability.

### **5.3 Problem Statement**

Although engineering students learn the theoretical foundations of data analytics, machine learning and artificial intelligence during their academic coursework, structured exposure to an end-to-end Data Science workflow is often limited. Important activities such as data engineering, data preparation, feature engineering, model validation, deployment, monitoring and platform administration require practical exposure beyond isolated model-building exercises.

The problem addressed by this internship was to practically learn and understand the complete RapidMiner-based Data Science workflow from identifying business problems and accessing and preparing data, to building and validating machine-learning models, and further understanding deployment, operationalization and platform administration.

### **5.4 Project Objectives**

- Understand the foundations and terminology of Data Science, Artificial Intelligence and Machine Learning.
- Identify business problems suitable for AI and machine-learning approaches.
- Understand and apply the CRISP-DM methodology to Data Science and machine-learning problems.
- Develop practical skills for accessing, loading, transforming, joining, aggregating and preparing data in RapidMiner.
- Understand supervised and unsupervised machine-learning techniques and correctly validate models.
- Develop advanced skills in data preparation, feature engineering, scripting and automation.

- Study advanced classification, regression, time-series handling, model selection, validation and optimization.
- Understand model deployment, monitoring and management considerations.
- Gain foundational knowledge of RapidMiner AI Hub administration, security, scalability, scheduling and user management.
- Complete the final credential validation / Internship Grade Point Assessment.

### **5.5 Scope of the Project**

The scope of this internship spans the complete Data Science workflow covered by the Siemens Data Science Master pathway. It includes business understanding and use-case identification, data access and engineering, data transformation and preparation, supervised and unsupervised learning, feature engineering, model validation and optimization, deployment and monitoring concepts, and RapidMiner AI Hub administration.

The internship was limited to a virtual, simulation/lab-based learning environment provided through the EduSkills and RapidMiner learning ecosystem, and did not involve deployment on any live organizational production system.

### **5.6 Technologies and Tools Used**

- Data Science Platform: RapidMiner Studio / RapidMiner Academy learning environment
- Data Engineering: Data loading, ETL, transformations, filtering, joins, append, pivot and aggregation
- Machine Learning: Classification, regression, clustering, association analysis, k-NN, decision trees, neural networks, Naive Bayes and GLM
- Data Preparation: Missing-value handling, type conversion, attribute generation and selection, feature engineering
- Advanced Data Handling: Loops, branches, variables, error handling, scripting and text-data processing
- Model Evaluation: Validation, scoring, model selection and optimization
- Deployment & Operations: Model deployment, monitoring, management and web applications
- Platform Administration: RapidMiner AI Hub installation/configuration, security, scalability, scheduling, users, licenses, extensions and connections

### **5.7 System Architecture**

The general architecture followed during the internship for the Siemens Data Science Master learning track can be summarized in the following layers:

- Business Layer: Identify a business problem and define the analytical objective.
- Data Layer: Access, load and understand data from relevant sources.
- Preparation Layer: Clean, transform, join, aggregate and enrich the data; prepare features.
- Model Layer: Train supervised or unsupervised machine-learning models using appropriate algorithms.
- Validation Layer: Validate, score, compare and optimize models.
- Deployment Layer: Understand deployment, monitoring and model-management workflows.
- Platform Layer: Understand AI Hub administration, security, scalability and scheduling.
- Assessment Layer: Weekly learning-path assessments, certifications and final credential validation track progress and grading.

## 5.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, practical learning-path activities and quiz-based assessments, culminating in a Final Credential Validation / Internship Grade Point Assessment, as summarized in the table below.

| Week  | Module                                | Description                                                                                                                                      |
|-------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Applications & Use Cases Professional | Learn to identify business problems suitable for AI and machine learning; understand Data Science terminology, CRISP-DM and analytics use cases. |
| 2     | Data Engineering Professional         | Learn to access, load, transform, join, append, pivot, aggregate and calculate data using RapidMiner.                                            |
| 3     | Machine Learning Professional         | Build foundational machine-learning skills including classification, regression, clustering, validation and model evaluation.                    |
| 4     | Data Preparation Master               | Master advanced data handling, preparation, feature engineering, automation and scripting.                                                       |
| 5     | Machine Learning Master               | Learn advanced machine-learning techniques, time-series handling, model selection, validation and optimization.                                  |
| 6     | Applications & Use Cases Master       | Understand the transition from a good model to production through deployment, monitoring and model management.                                   |
| 7     | Platform Administration Master        | Configure and manage RapidMiner AI Hub, including security, scalability, scheduling and user management.                                         |
| 8     | Platform Administration Master        | Develop enterprise-level understanding of platform operations, configuration, security and administration.                                       |
| Final | Final Credential Validation           | Final Credential Validation / Internship Grade Point Assessment.                                                                                 |

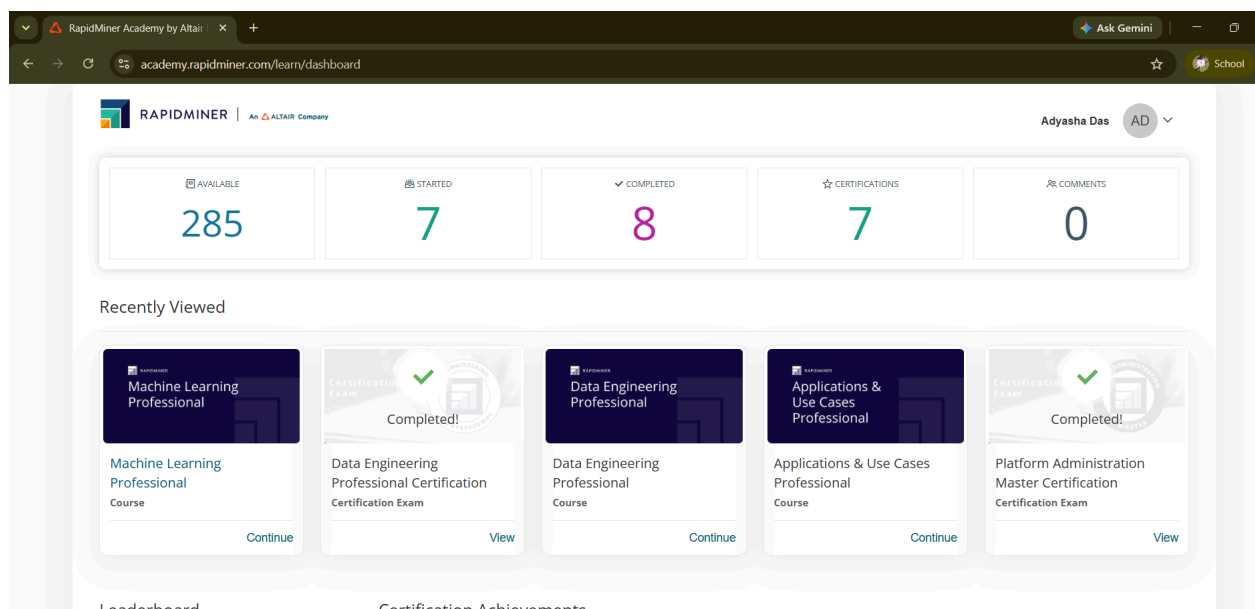
Each week concluded with learning-pathway assessments and certification progress activities to evaluate conceptual understanding. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the internship requirements and final assessment.

## 5.9 Expected Outcomes

- Ability to explain and apply Data Science concepts and identify suitable machine-learning use cases.
- Practical understanding of data engineering and data preparation workflows using RapidMiner.
- Understanding of supervised and unsupervised machine-learning techniques and model validation.
- Exposure to advanced feature engineering, scripting, time-series handling, model selection and optimization.
- Understanding of model deployment, monitoring and model-management considerations.
- Foundational knowledge of RapidMiner AI Hub platform administration.
- A completed set of learning-path modules and assessments, along with successful clearance of the Final Credential Validation / Internship Grade Point Assessment, resulting in an “Outstanding” grade.

## 5.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Siemens Data Science Master, are attached in this report. The supporting screenshots below provide evidence of the RapidMiner learning-path progress, course completion and related internship work.



Leaderboard

Certification Achievements

YOUR STANDING

145 YOU

700 RAPIDMINER NUGGETS

1 LEADER

800 RAPIDMINER NUGGETS

CERTIFICATION GRAND MASTER

700 RAPIDMINER NUGGETS

CERTIFICATION MASTER

600 RAPIDMINER NUGGETS

CERTIFICATION ADEPT

500 RAPIDMINER NUGGETS

CERTIFICATION KNIGHT

400 RAPIDMINER NUGGETS

CERTIFICATION PADAWAN

300 RAPIDMINER NUGGETS

CERTIFICATION ACO

200 RAPIDMINER NUGGETS

Activity

My Learning 7

Completed 8

Certifications 7

|                                                       |                    |                                    |
|-------------------------------------------------------|--------------------|------------------------------------|
| > Data Science Professional with RapidMiner           | Learning Path      | <a href="#">View Learning Path</a> |
| > Applications & Use Cases Professional Certification | Certification Exam | <a href="#">View</a>               |
| > Data Engineering Professional Certification         | Certification Exam | <a href="#">View</a>               |
| > Machine Learning Professional Certification         | Certification Exam | <a href="#">View</a>               |
| > Data Engineering Master Certification               | Certification Exam | <a href="#">View</a>               |
| > Machine Learning Master Certification               | Certification Exam | <a href="#">View</a>               |

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Labels

AICTE-EduSkills Internship 2024 Certificate download link

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2 of 5

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to me

Mon, Aug 10, 2:36 PM

Star

Smiley

Reply

More

Dear Adyasha Das,

Congratulations!

You have successfully completed 10 weeks Siemens Data Science Master internship with EduSkills Foundation. Your certificate is ready — please log in to your profile and download it.

Steps to Download Your Certificate:

1. Login at: <https://aicte-internship.eduskillsfoundation.org/>

2. Go to the **Certificate Section** from your sidebar

3. Click **Download Certificate**

Please post your certificate on your **LinkedIn profile** to get better job opportunities and visibility from corporates.

Reply

Forward

Share in chat

Page 12

## **6. BIBLIOGRAPHY/REFERENCES**

- EduSkills, “Data Science Virtual Internship Offer Letter,” Ref. No. C17Y2026M081497680, 2026.
- EduSkills / AICTE, “Certificate of Virtual Internship — Data Science Master Virtual Internship,” 2026.
- RapidMiner Academy, “Data Science Professional with RapidMiner,” including Applications & Use Cases Professional, Data Engineering Professional and Machine Learning Professional learning materials.
- RapidMiner Academy, “Data Science Master with RapidMiner,” including Data Engineering Master, Machine Learning Master, Applications & Use Cases Master and Platform Administration Master learning materials.
- RapidMiner Academy, Applications & Use Cases Professional course materials.
- RapidMiner Academy, Data Engineering Professional course materials.
- RapidMiner Academy, Machine Learning Professional course materials.
- RapidMiner Academy, Data Science Master learning-path materials.